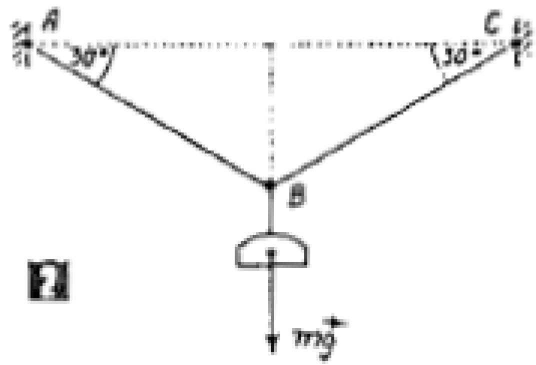
Problem 2

A load with mass $m = 100$ kg is hanging from a wire system ABC, where points A and C are fixed, and the angles with the horizontal at points A and C are $30°$. Find the tension forces in the segments AB and BC. Refer to the figure below:



Solution

The load exerts a downward force due to gravity:

$$F_g = mg = 100 \times 9.81 = 981 \text{ N}$$

Since the system is in equilibrium, the vertical and horizontal components of the tension forces must balance.

Let T_1 be the tension in segment AB, and T_2 be the tension in segment BC.

Horizontal equilibrium:

$$T_1 \cos(30°) = T_2 \cos(30°)$$

This implies:

$$T_1 = T_2$$

Vertical equilibrium:

$$T_1 \sin(30°) + T_2 \sin(30°) = mg$$

Substitute $T_1 = T_2$:

$$2T_1 \sin(30°) = 981$$

Since $\sin(30°) = 0.5$:

$$2T_1(0.5) = 981 T_1 = T_2 = \frac{981}{1} = 981 \text{ N}$$

Result: The tension forces in both segments are:

$$T_1 = T_2 = 981 \text{ N}$$

The given solution is correct!